

Moller Polarimetry Systematic Studies March 15, 2023

Hall-A Weekly Meeting March 28th 2023



TEMPLE
UNIVERSITY

**Jefferson Lab**
Exploring the Nature of Matter

Møller Systematic Studies

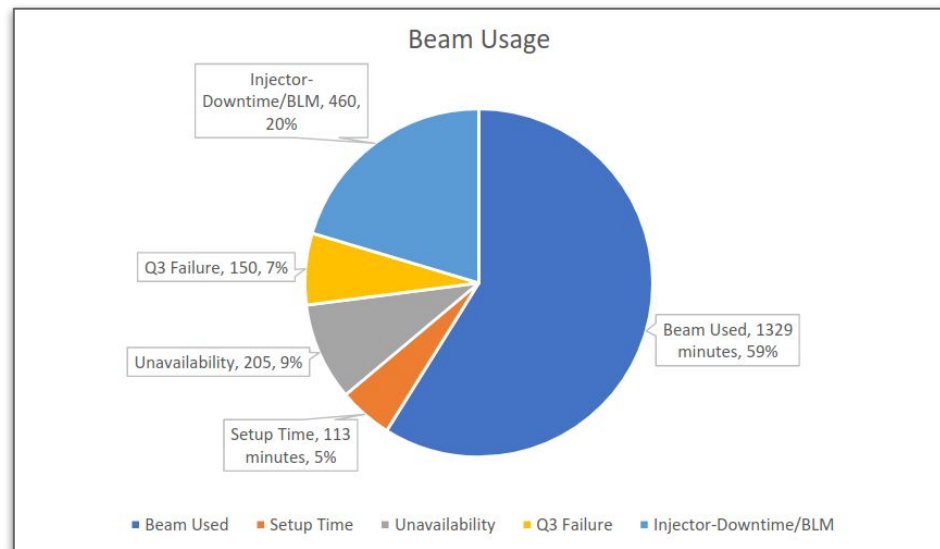
High-precision systematic studies performed at a precision beyond what's important to SBS:

- Target rotation study
 - Target field study
- Saturation
Studies
- Asymmetry scan at high-energy
 - This is a fundamentally good opportunity to check our analyzing power calculations at higher-energy.

Møller Measurement Summary

High-precision systematic studies performed at a precision beyond what's important to SBS:

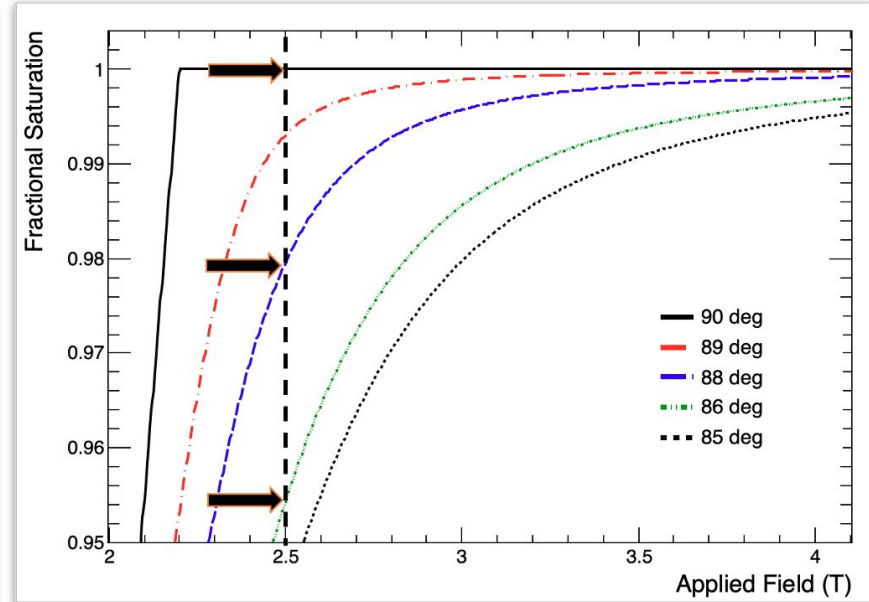
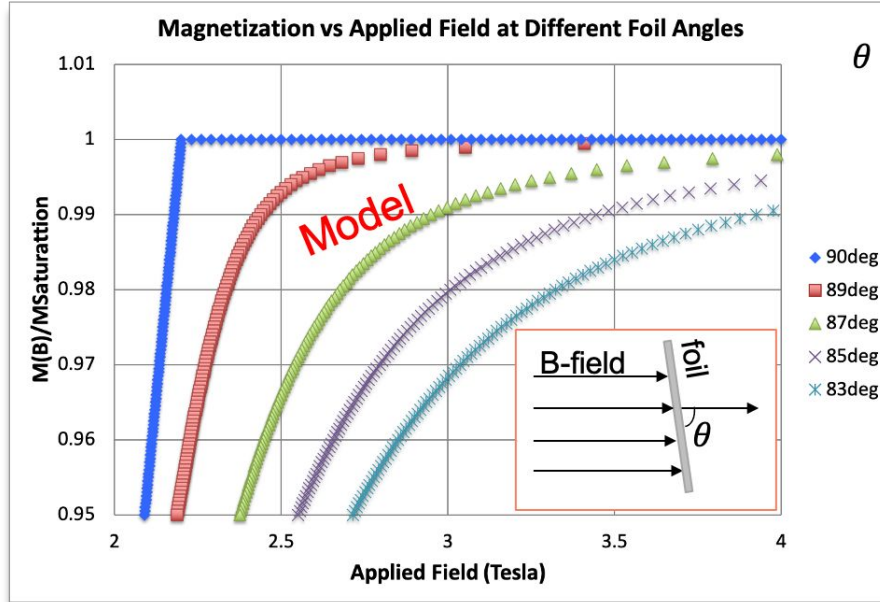
- Target rotation study
 - Target field study
 - Asymmetry scan at high-energy
 - This is a fundamentally good opportunity to check our analyzing power calculations at higher-energy.
- Saturation Studies



→ **Significant amount of available beam:**

~23 hours total of usable high-precision
Moller polarimetry data

Rotation and Saturation – Stoner-Wohlfarth Model



Saturation polarization of the target foil is highly sensitive to the alignment of the target.

⇒ Foil must be aligned perpendicularly to the field (90° deg) in order to saturate at 2.5T.

⇒ In practice, we use 4.0T field.

Rotation and Saturation – Stoner-Wohlfarth Model

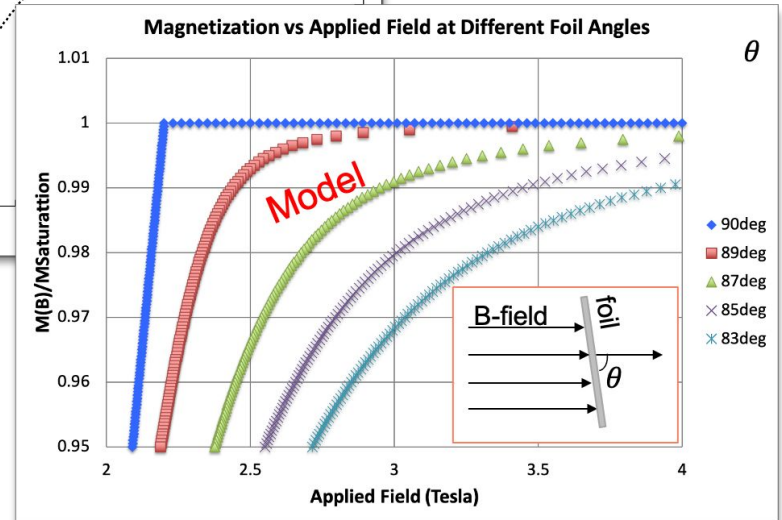
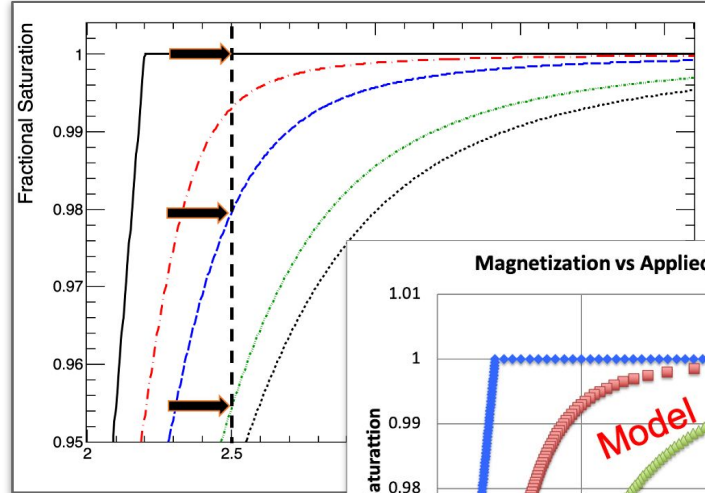
We need to accurately understand the uncertainty on the ‘fractional’ saturation.

Systematic uncertainty on the saturation polarization is 0.24%



[10.1016/j.nima.2022.167444](https://doi.org/10.1016/j.nima.2022.167444)

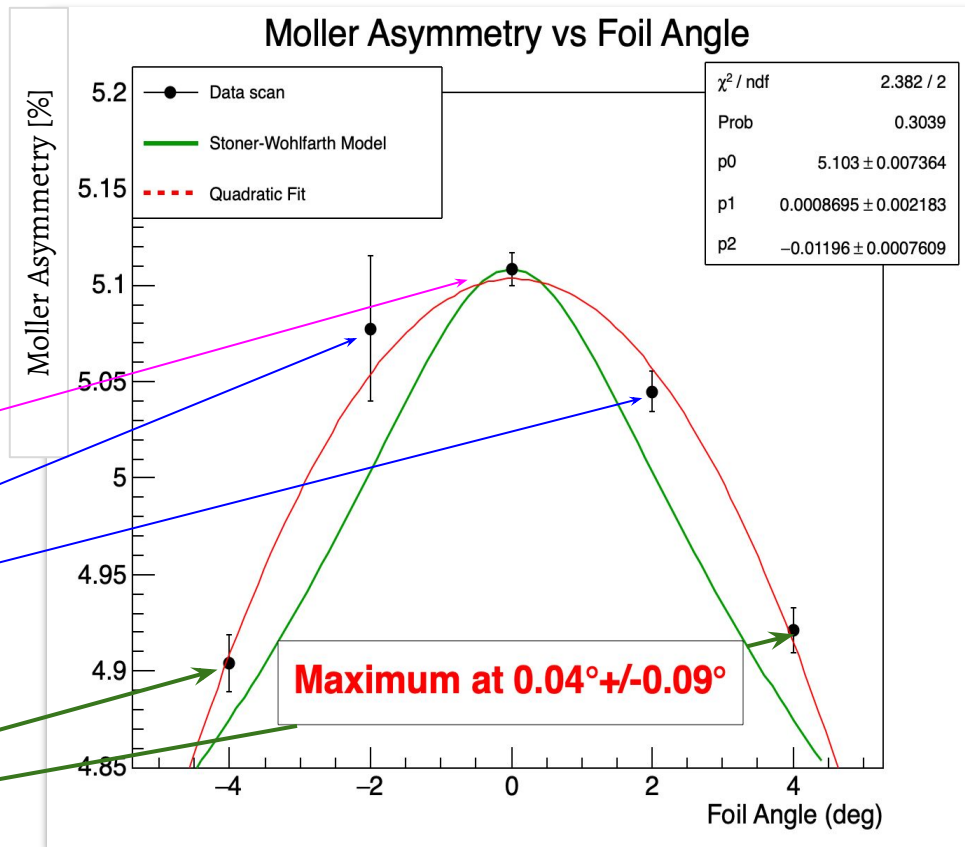
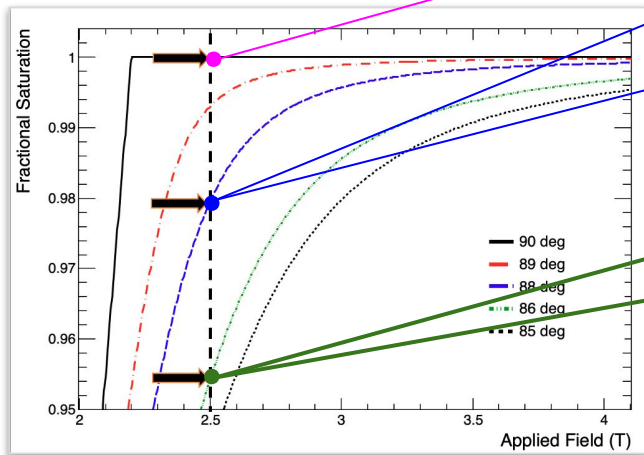
⇒ **Total target polarization error budget is 0.28% ⇐**



Foil Rotation

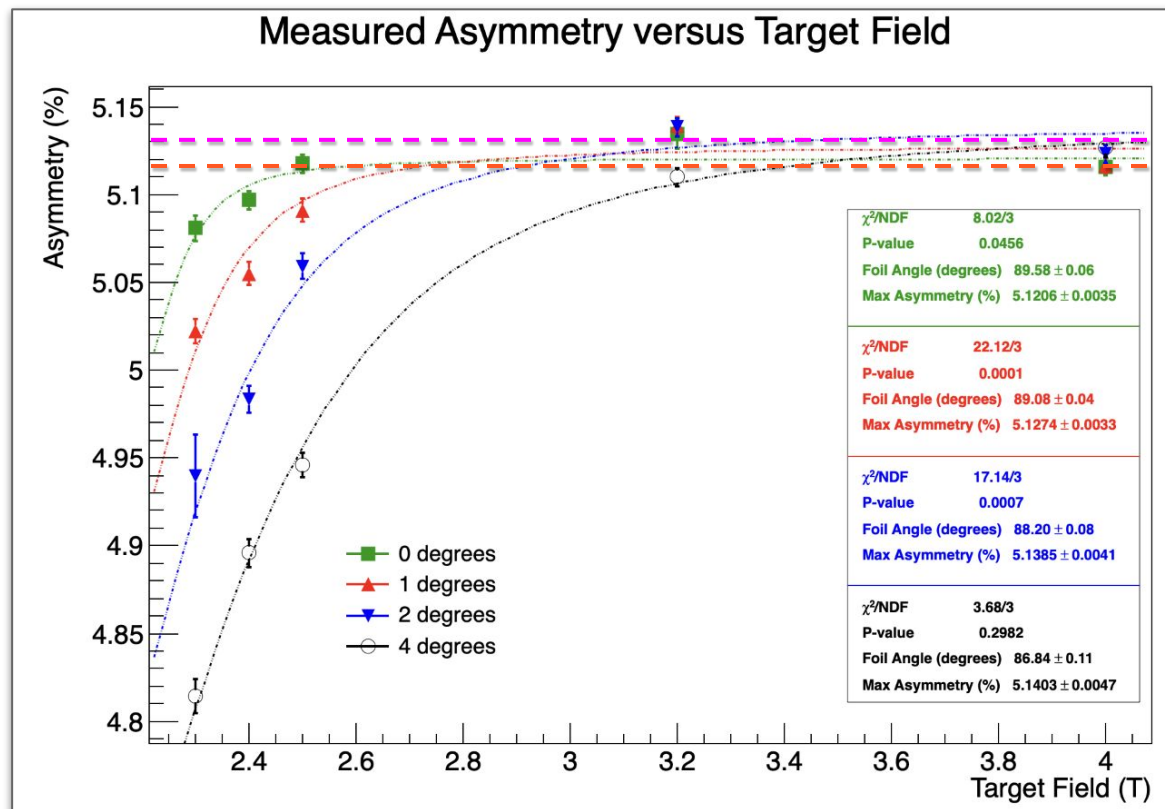
Moller foil rotation re-enabled

- Was locked after the PREX2 Commissioning in March 2019
- Second-order poly fit parameters shown.
 - Fit consistent with foil being perpendicular to beam



Moller Target Saturation Scan

- Saturation scans performed ramping the target-holding field.
- Results compared to Stoner curves
 - Results look okay.
 - These were taken collectively over a long period of time.



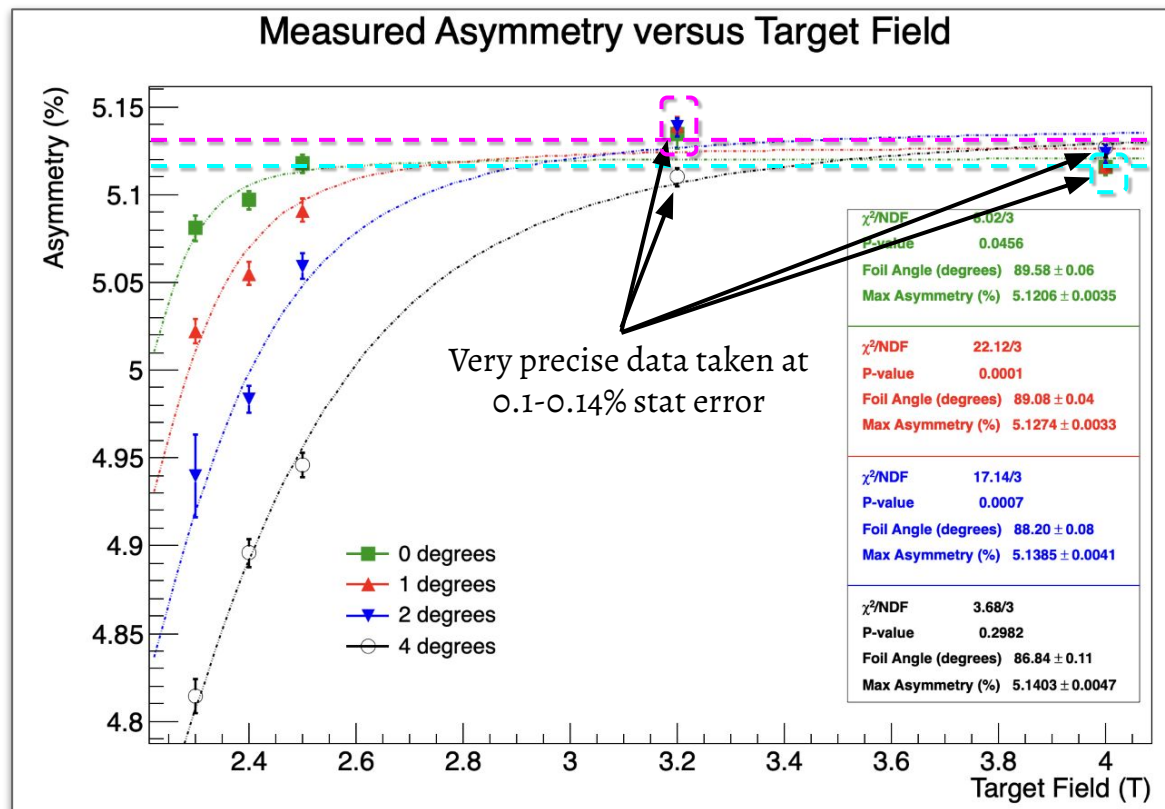
Moller Target Saturation Scan

Transient beam polarization wandering between 4.0T and 3.2T measurements.

- Asymmetry:
 $0.05135 \leftrightarrow 0.0512$
- **~0.3% difference**

Downloaded some data from the EPICs archive

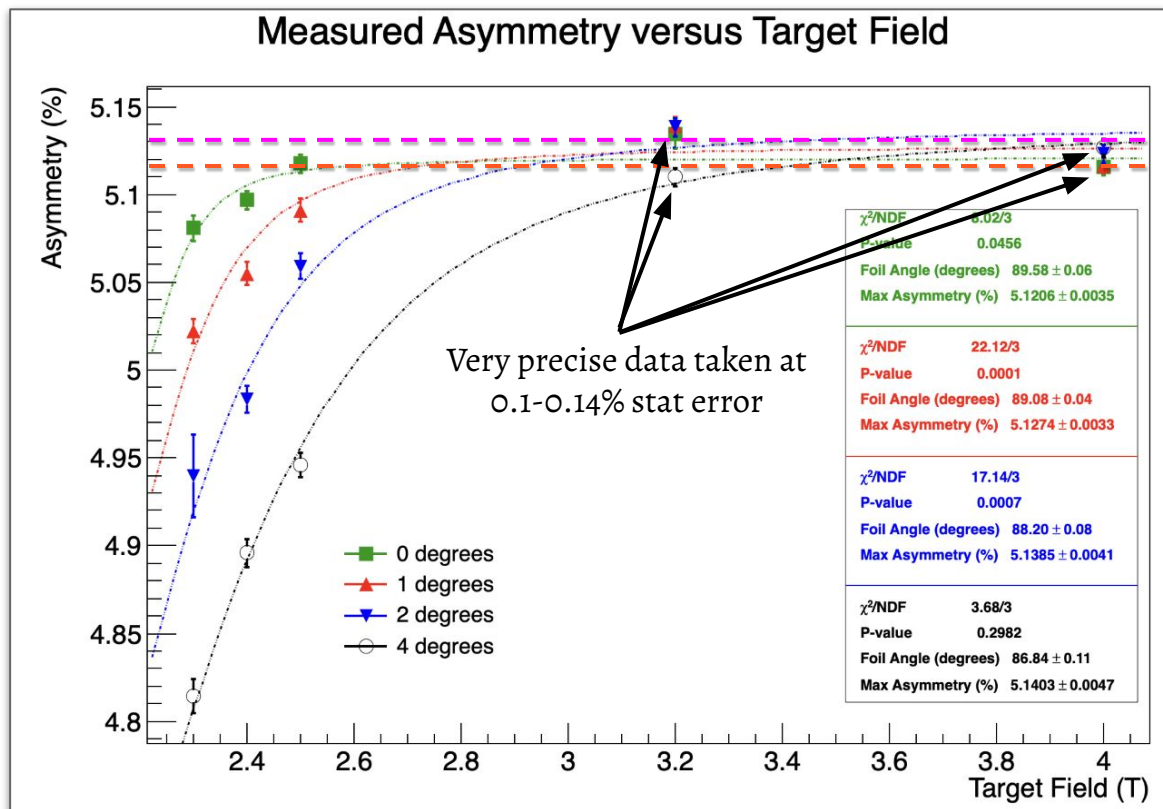
- Look for correlations and try a correction for precession
- Hall A was sharing polarization which could leave us sensitive to momentum changes.



Moller Target Saturation Scan

Hall A was sharing polarization which could leave us sensitive to momentum changes.

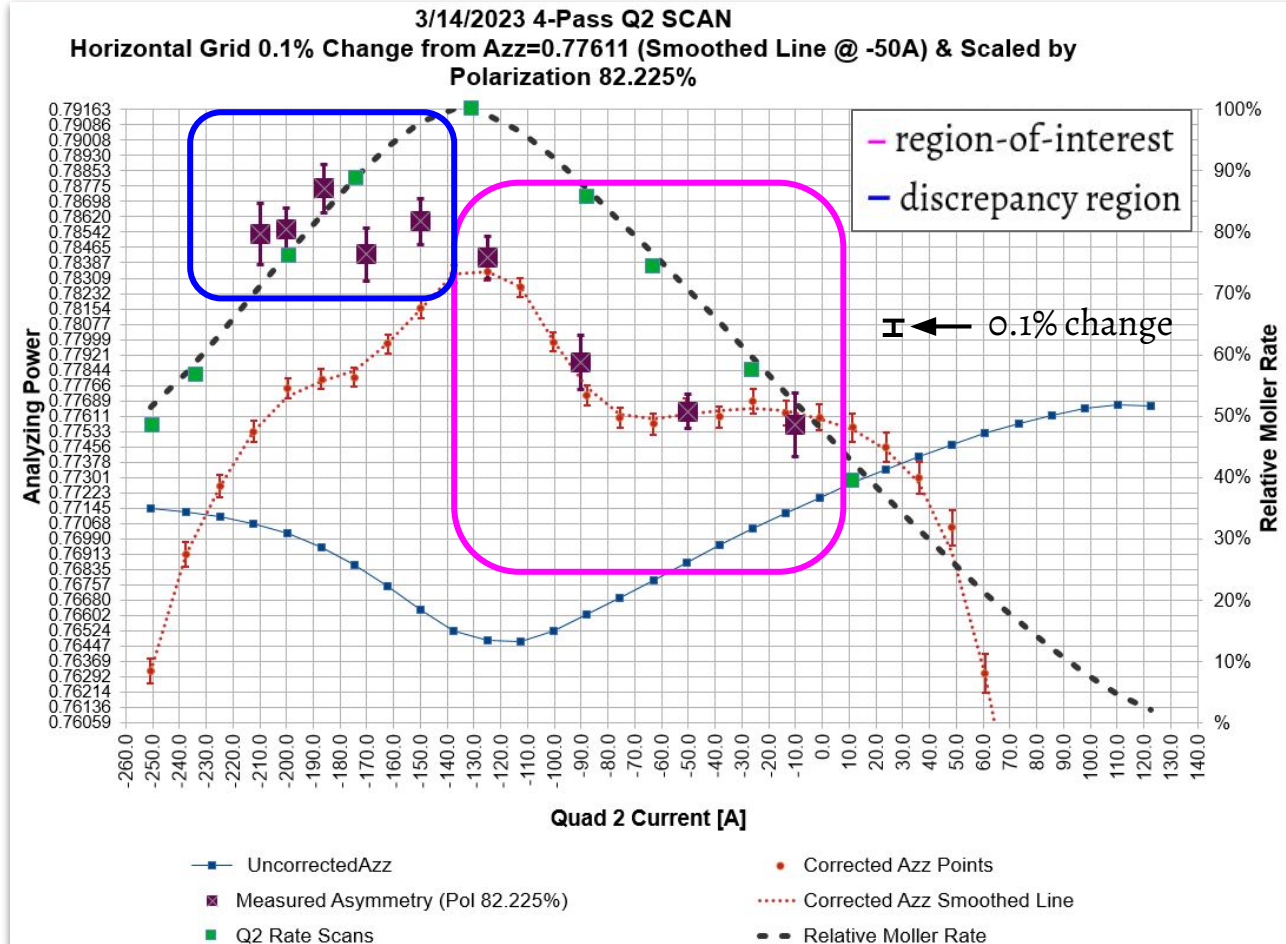
- Setup at 17* deg off longitudinal **AND** at 4-Pass energy makes us sensitive to precession.
- Between the Injector and the Hall this translates to 13,000 degrees of precession.
- A $\sim 0.005\%$ change in momentum results $\sim 0.33\%$ difference in polarization.



Asymmetry Scan

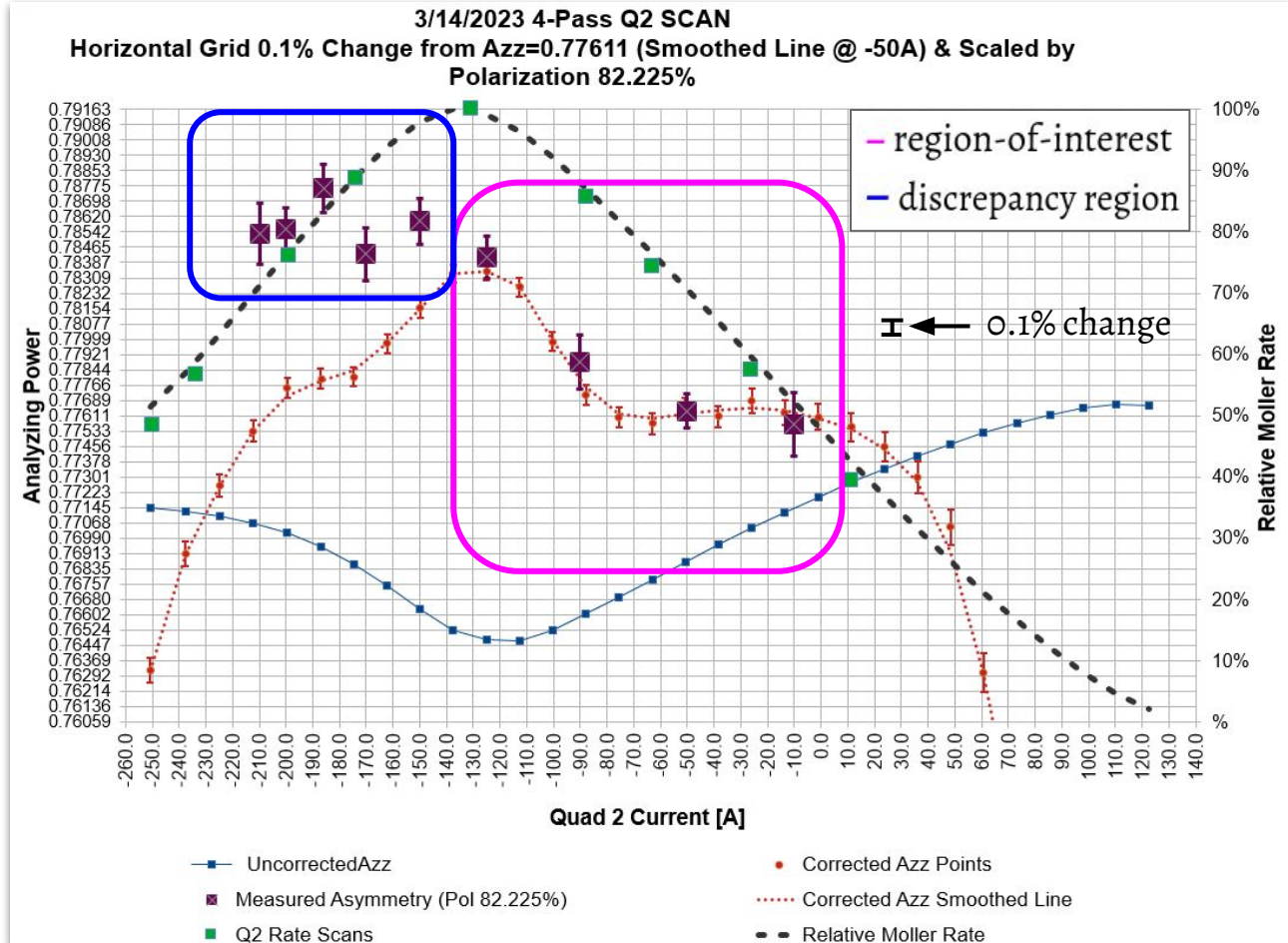
Points about this plot:

- 1) Black dashed line is the Moller rate normalized to maximum.
- 2) Region of interest is where our polarization measurements generally take place.
- 3) Discrepancy region... where the discrepancy occurs; I'll discuss this.



Asymmetry Scan

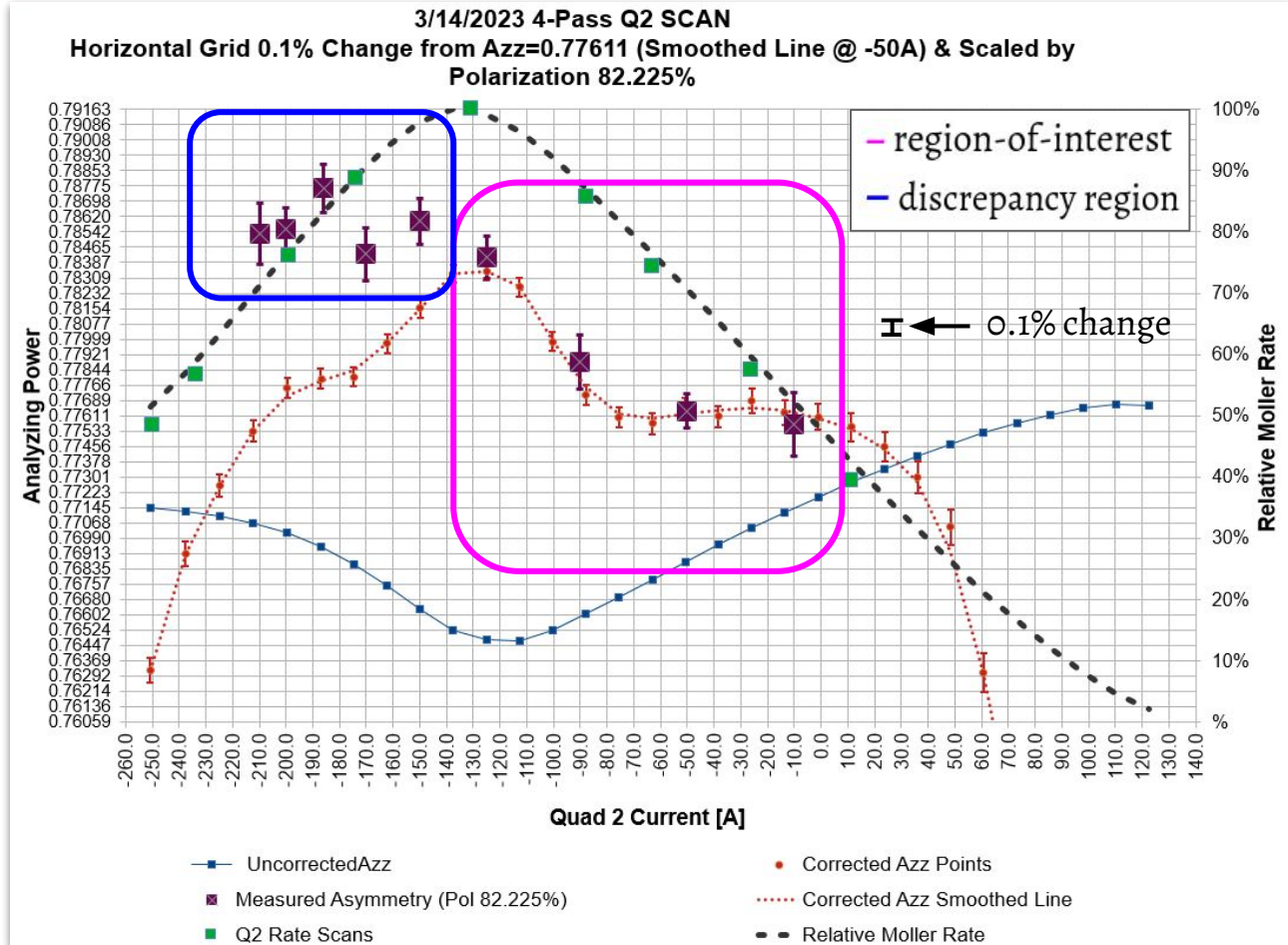
- First high-energy high-precision asymmetry scan that we have.
- Data fits well in our region of interest
 - Within statistical error.



Asymmetry Scan

Asymmetry correction differs in discrepancy region.

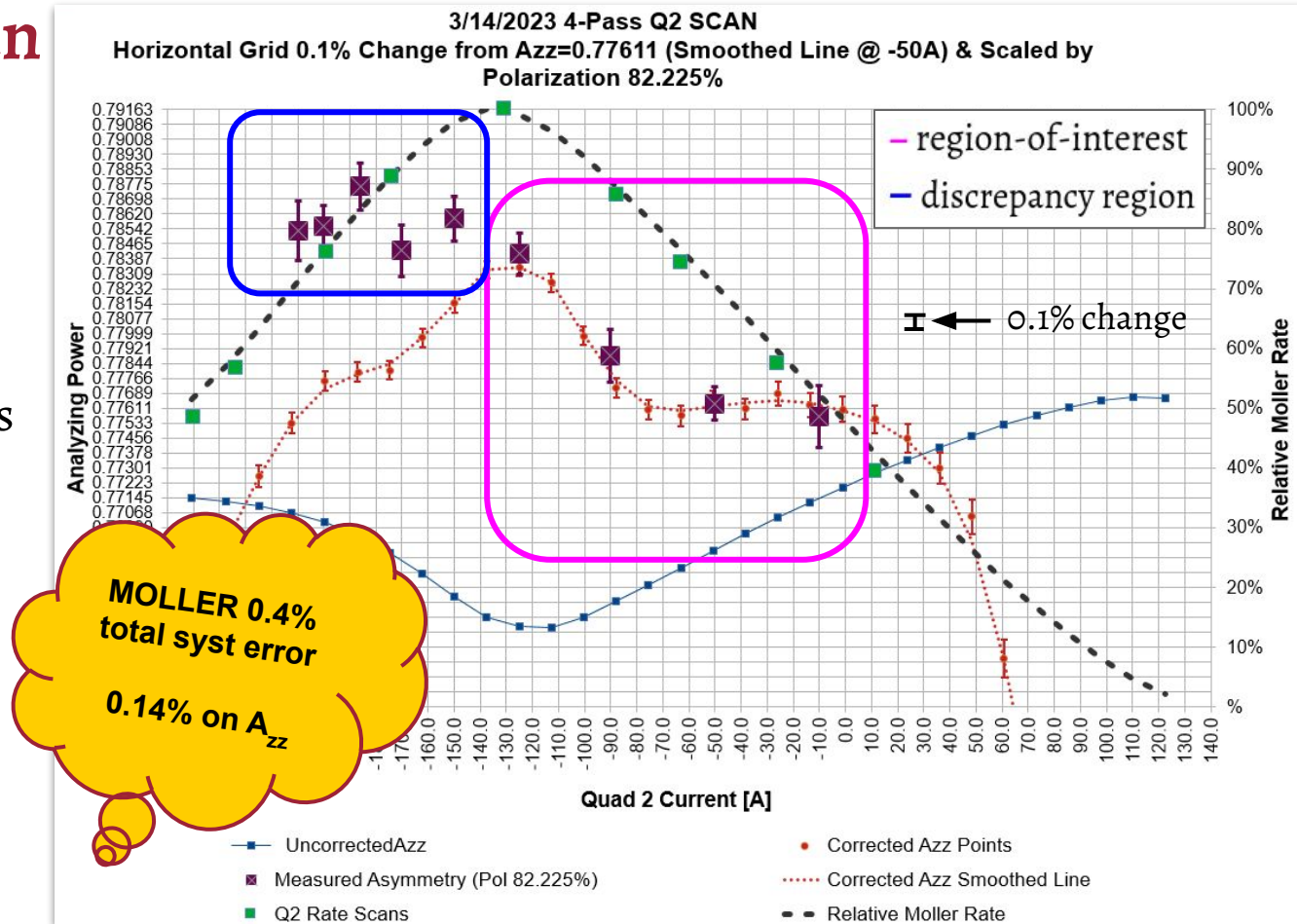
- Re-simulating using TOSCA field maps.
- High-momentum electrons pass close to beam pipe.
- Field gradient changes and transportation will differ from ideal fields.



Asymmetry Scan

If we nail this down then analyzing power calculations insofar as Levchuk and transportation becomes settled.

►►►► The Q3 power supply breaking may have been a blessing in disguise





Ongoing Data Analysis



- Analyzing power calculations being re-performed using TOSCA field maps:
 - ➡ Does this do a better job at explaining difference in asymmetry when Q2 was heavily defocusing?
- Saturation scan data going to be analyzed to see what may have caused drifts:
 - ➡ Correction for precession needed?
 - ➡ Other causes?



The good news is that we have quality well-controlled data.